Standard Model

1. Spontaneous Symmetry Breaking

Spontaneous symmetry breaking is the situation in which the equations and the Lagrangian have a symmetry, but the vacuum chosen by the theory does not. The symmetry is still present in the theory; it is not a mistake in the Lagrangian. The point is that the set of lowest-energy field configurations contains more than one equivalent choice, and choosing one of them hides part of the symmetry.

This section develops the idea in three steps. First we study a real scalar with a discrete symmetry, where there is symmetry breaking but no Goldstone boson. Then we study a complex scalar with a continuous $U(1)$ symmetry, where a massless Goldstone mode appears. Finally we generalize to N complex scalars and prove the Goldstone theorem by counting flat directions of the potential.

1.1 A real scalar with a discrete symmetry

Consider one real scalar field ϕ with Lagrangian

$$\mathcal{L} = \frac{1}{2}\partial_\mu\phi\,\partial^\mu\phi - V(\phi), \tag{4.1}$$

and potential

$$V(\phi) = -\frac{1}{2}\mu^2\phi^2 + \frac{\lambda}{4}\phi^4, \qquad \mu^2 > 0, \quad \lambda > 0. \tag{4.2}$$

This is often called the "wrong-sign mass" potential. Around $\phi = 0$ the quadratic term in the potential has negative curvature, so $\phi = 0$ is not a stable vacuum. This corrects a common misleading phrase: the theory does have a quadratic term; what is wrong is the sign of the quadratic term around the origin.

The potential is invariant under the discrete transformation

$$\phi \longrightarrow -\phi, \qquad V(\phi) = V(-\phi). \tag{4.3}$$

This is a $\mathbb{Z}_2$ symmetry.

To find the extrema, compute

$$\frac{\mathrm{d}V}{\mathrm{d}\phi} = -\mu^2\phi + \lambda\phi^3 = \phi\left(-\mu^2 + \lambda\phi^2\right). \tag{4.4}$$

Thus

$$\frac{\mathrm{d}V}{\mathrm{d}\phi} = 0 \qquad \Longrightarrow \qquad \phi = 0 \quad \text{or} \quad \phi^2 = \frac{\mu^2}{\lambda}. \tag{4.5}$$

Define

$$v = \frac{\mu}{\sqrt{\lambda}}. \tag{4.6}$$

The extrema are therefore

$$\phi = 0, \qquad \phi = \pm v. \tag{4.7}$$

To decide which are minima, compute the second derivative:

$$\frac{\mathrm{d}^2 V}{\mathrm{d}\phi^2} = -\mu^2 + 3\lambda\phi^2. \tag{4.8}$$

At the origin,

$$\left.\frac{\mathrm{d}^2 V}{\mathrm{d}\phi^2}\right|_{\phi=0} = -\mu^2 < 0, \tag{4.9}$$

so $\phi = 0$ is a local maximum. At $\phi = \pm v$,

$$\left.\frac{\mathrm{d}^2 V}{\mathrm{d}\phi^2}\right|_{\phi=\pm v} = -\mu^2 + 3\lambda\frac{\mu^2}{\lambda} = 2\mu^2 > 0, \tag{4.10}$$

so $\phi = \pm v$ are true minima.

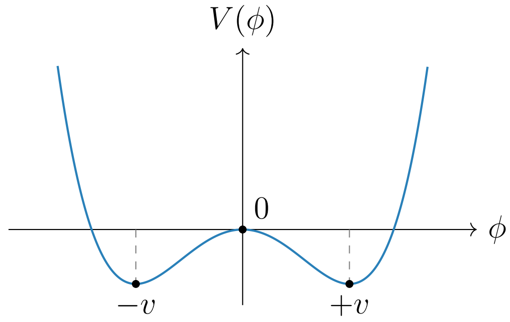

Figure 4.1: The double-well potential. The Lagrangian is symmetric under $\phi \to -\phi$, but either vacuum $\phi = +v$ or $\phi = -v$ chooses one side.

The theory has the symmetry $\phi \to -\phi$, but the vacuum does not. If the system chooses

$$\langle 0|\phi|0\rangle = +v, \tag{4.11}$$

then the transformed configuration has

$$\langle 0|\phi|0\rangle \longrightarrow -v, \tag{4.12}$$

which is a different vacuum. The two vacua are physically equivalent, but once one is chosen the symmetry is no longer manifest. This is spontaneous breaking of a discrete symmetry.

There is no Goldstone boson in this example. Goldstone bosons arise from broken continuous symmetries, where moving along the vacuum manifold can be done infinitesimally with no energy cost. A discrete symmetry does not provide such a continuous flat direction.

1.2 A complex scalar and global U(1) breaking

Now consider a complex scalar field

$$\phi = \frac{1}{\sqrt{2}}\left(\phi_1 + \mathrm{i}\phi_2\right), \tag{4.13}$$

with Lagrangian

$$\mathcal{L} = \partial_\mu\phi^*\,\partial^\mu\phi - V(\phi, \phi^*), \tag{4.14}$$

and potential

$$V(\phi,\phi^*) = -\mu^2\phi^*\phi + \frac{\lambda}{2}\left(\phi^*\phi\right)^2, \qquad \mu^2>0, \quad \lambda>0. \tag{4.15}$$

The Lagrangian is invariant under the global $U(1)$ transformation

$$\phi(x) \longrightarrow \phi'(x) = e^{\mathrm{i}\alpha}\phi(x), \qquad \alpha\in\mathbb{R}. \tag{4.16}$$

This is a global symmetry because α is constant in spacetime.

The potential depends only on the radial variable $\phi^*\phi$. Minimizing with respect to ϕ^* gives

$$\frac{\partial V}{\partial\phi^*} = \phi\left(-\mu^2+\lambda\phi^*\phi\right) = 0. \tag{4.17}$$

Therefore

$$\phi = 0 \quad \text{or} \quad \phi^*\phi = \frac{\mu^2}{\lambda}. \tag{4.18}$$

Again $\phi=0$ is the unstable point. The true vacua satisfy

$$|\phi|^2 = \frac{\mu^2}{\lambda}. \tag{4.19}$$

In the complex plane this is a circle. Since

$$|\phi|^2 = \frac{1}{2}\left(\phi_1^2+\phi_2^2\right), \tag{4.20}$$

the vacuum circle can also be written as

$$\phi_1^2+\phi_2^2 = \frac{2\mu^2}{\lambda}. \tag{4.21}$$

It is conventional to define

$$v^2 = \frac{2\mu^2}{\lambda}. \tag{4.22}$$

Then the vacuum condition is

$$|\phi|^2 = \frac{v^2}{2}. \tag{4.23}$$

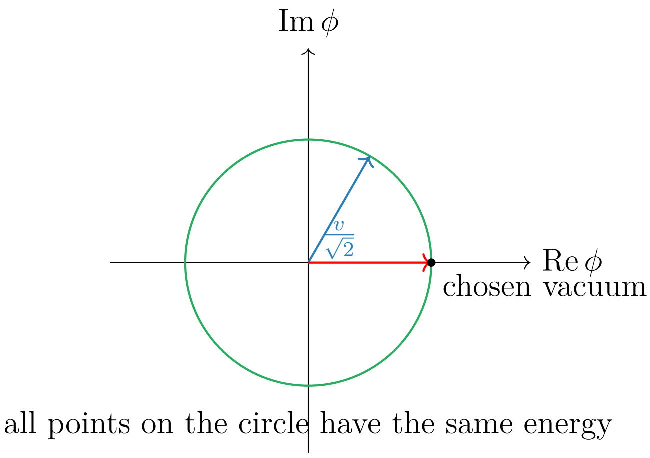

Figure 4.2: The vacuum manifold for the broken global $U(1)$ model. Choosing one point on the circle breaks the manifest $U(1)$ symmetry.

Every point on this circle is a possible ground state. Choosing one point breaks the global $U(1)$ symmetry. We may choose the vacuum to lie along the real direction:

$$\phi_0 = \frac{v}{\sqrt{2}}, \qquad \phi_{1,0} = v, \quad \phi_{2,0} = 0. \tag{4.24}$$

This choice is not a loss of generality; any other point on the circle can be rotated into this one by a $U(1)$ transformation.

1.3 Expansion around the chosen vacuum

To find the physical particles, expand the field around the chosen vacuum:

$$\phi(x) = \frac{1}{\sqrt{2}} [v + h(x) + \mathrm{i}\sigma(x)] . \tag{4.25}$$

Here $h(x)$ is the radial fluctuation and $\sigma(x)$ is the angular fluctuation. At the vacuum,

$$h = 0, \qquad \sigma = 0. \tag{4.26}$$

The kinetic term becomes

$$\begin{aligned} \partial_\mu \phi^* \, \partial^\mu \phi &= \frac{1}{2} \partial_\mu (v + h - \mathrm{i}\sigma) \partial^\mu (v + h + \mathrm{i}\sigma) \\ &= \frac{1}{2} \partial_\mu h \, \partial^\mu h + \frac{1}{2} \partial_\mu \sigma \, \partial^\mu \sigma. \end{aligned} \tag{4.27}$$

Since v is constant, its derivative vanishes.

Now compute the potential in terms of h and σ. First,

$$\begin{aligned} \phi^* \phi &= \frac{1}{2} \left[(v + h)^2 + \sigma^2 \right] \\ &= \frac{1}{2} \left[v^2 + 2vh + h^2 + \sigma^2 \right] . \end{aligned} \tag{4.28}$$

Therefore

$$V(h, \sigma) = -\frac{\mu^2}{2} \left[(v + h)^2 + \sigma^2 \right] + \frac{\lambda}{8} \left[(v + h)^2 + \sigma^2 \right]^2 . \tag{4.29}$$

Expanding the square,

$$\begin{aligned} \left[(v + h)^2 + \sigma^2 \right]^2 &= \left(v^2 + 2vh + h^2 + \sigma^2 \right)^2 \\ &= v^4 + 4v^3 h + 6v^2 h^2 + 2v^2 \sigma^2 + \mathcal{O}(\text{cubic}). \end{aligned} \tag{4.30}$$

Hence the terms up to quadratic order are

$$V(h, \sigma) = -\frac{\mu^2}{2} \left(v^2 + 2vh + h^2 + \sigma^2 \right) + \frac{\lambda}{8} \left(v^4 + 4v^3 h + 6v^2 h^2 + 2v^2 \sigma^2 \right) + \cdots . \tag{4.31}$$

Separate constant, linear, and quadratic terms:

$$\begin{aligned} V(h, \sigma) = &\left(-\frac{\mu^2 v^2}{2} + \frac{\lambda v^4}{8} \right) + \left(-\mu^2 v + \frac{\lambda v^3}{2} \right) h \\ &+ \left(-\frac{\mu^2}{2} + \frac{3\lambda v^2}{4} \right) h^2 + \left(-\frac{\mu^2}{2} + \frac{\lambda v^2}{4} \right) \sigma^2 + \cdots . \end{aligned} \tag{4.32}$$

Using

$$v^2 = \frac{2\mu^2}{\lambda}, \tag{4.33}$$

the linear term vanishes:

$$-\mu^2 v + \frac{\lambda v^3}{2} = v \left(-\mu^2 + \frac{\lambda v^2}{2} \right) = 0. \tag{4.34}$$

This must happen because we expanded around a minimum. The quadratic terms become

$$-\frac{\mu^2}{2}+\frac{3\lambda v^2}{4}=-\frac{\mu^2}{2}+\frac{3\mu^2}{2}=\mu^2, \tag{4.35}$$
$$-\frac{\mu^2}{2}+\frac{\lambda v^2}{4}=-\frac{\mu^2}{2}+\frac{\mu^2}{2}=0. \tag{4.36}$$

Thus

$$V(h,\sigma)=V(v)+\mu^2h^2+\mathcal{O}(h^3,h^2\sigma,h\sigma^2,\sigma^4). \tag{4.37}$$

The Lagrangian contains $-\mu^2h^2$. Comparing with the standard mass term

$$-\frac{1}{2}m_h^2h^2, \tag{4.38}$$

we find

$$m_h^2=2\mu^2. \tag{4.39}$$

There is no quadratic term for σ, so

$$m_\sigma^2=0. \tag{4.40}$$

The radial field h is massive. The angular field σ is massless and is called the Goldstone boson.

The reason is geometric. The h direction moves away from the vacuum circle, so the potential increases. The σ direction moves along the vacuum circle, where the potential is flat. A flat direction has zero curvature, and zero curvature means zero mass squared.

1.4 N complex scalar fields

Now consider N complex scalar fields ϕ^i, $i=1,\dots,N$, with

$$\mathcal{L}=\sum_{i=1}^{N}\partial_\mu\phi^{i*}\partial^\mu\phi^i-V(\phi,\phi^*), \tag{4.41}$$

and

$$V(\phi,\phi^*)=-\mu^2\sum_{i=1}^{N}\phi^{i*}\phi^i+\frac{\lambda}{2}\left(\sum_{i=1}^{N}\phi^{i*}\phi^i\right)^2. \tag{4.42}$$

This potential is invariant under global $U(N)$ transformations:

$$\phi^i\longrightarrow\phi'^i=\sum_{j=1}^{N}U^{ij}\phi^j,\qquad U^\dagger U=\mathbf{1}. \tag{4.43}$$

The quantity

$$\sum_i\phi^{i*}\phi^i \tag{4.44}$$

is just the squared length of the complex vector ϕ, so it is unchanged by unitary rotations.

The minimum satisfies

$$\sum_{i=1}^{N}\phi^{i*}\phi^i=\frac{\mu^2}{\lambda}. \tag{4.45}$$

This is a sphere in the $2N$ real-dimensional field space. Define again

$$\frac{v^2}{2}=\frac{\mu^2}{\lambda},\qquad v^2=\frac{2\mu^2}{\lambda}. \tag{4.46}$$

Using the $U(N)$ symmetry, choose the vacuum to point in the first complex direction:

$$\phi_0^1 = \frac{v}{\sqrt{2}}, \qquad \phi_0^2 = \cdots = \phi_0^N = 0. \tag{4.47}$$

Then write the fields as

$$\phi^1 = \frac{1}{\sqrt{2}}\left[v + h(x) + \mathrm{i}\sigma(x)\right], \tag{4.48}$$

$$\phi^k = \frac{1}{\sqrt{2}}\left[\pi_{2k-2}(x) + \mathrm{i}\pi_{2k-1}(x)\right], \qquad k = 2, \dots, N. \tag{4.49}$$

There are $2N$ real fields in total:

$$h, \quad \sigma, \quad \pi_2, \dots, \pi_{2N-1}. \tag{4.50}$$

The radial field h changes the length of the vector ϕ and is massive. All other fields change only the direction of ϕ in the vacuum manifold and are massless at tree level.

Equivalently, the chosen vacuum leaves a residual $U(N-1)$ symmetry acting on the fields

$$\phi^2, \dots, \phi^N. \tag{4.51}$$

Thus the symmetry-breaking pattern is

$$U(N) \longrightarrow U(N-1). \tag{4.52}$$

The number of broken generators is

$$\dim U(N) - \dim U(N-1) = N^2 - (N-1)^2 = 2N - 1. \tag{4.53}$$

Therefore there are

$$2N - 1 \tag{4.54}$$

massless Goldstone bosons. This agrees with the field decomposition above: σ plus the $2N-2$ real components of $\phi^2, \dots, \phi^N$.

1.5 Goldstone theorem

The examples above illustrate the general result.

> **Goldstone theorem.** In a relativistic theory with a continuous global internal symmetry, each spontaneously broken generator gives one massless real scalar excitation, called a Goldstone boson.

There are important assumptions in this statement. The symmetry must be global, not gauged. If the symmetry is gauged, the massless scalar is removed from the physical spectrum by the Higgs mechanism. Also, the symmetry must be exact. If it is explicitly broken, the would-be Goldstone boson generally becomes light but not massless.

1.6 Proof by the mass matrix

Consider N real scalar fields ϕ^i with Lagrangian

$$\mathcal{L} = \frac{1}{2}\sum_{i=1}^{N} \partial_\mu \phi^i \, \partial^\mu \phi^i - V(\phi). \tag{4.55}$$

Suppose the potential is invariant under a continuous symmetry group G. For an infinitesimal transformation,

$$\phi^i \longrightarrow \phi^i + \delta_a \phi^i, \tag{4.56}$$

where

$$\delta_a \phi^i = \alpha^a (T_a)^i{}_j \phi^j. \tag{4.57}$$

Here T_a is a generator of the symmetry and α^a is an infinitesimal real parameter. Since V is invariant,

$$\delta_a V = \sum_i \frac{\partial V}{\partial \phi^i} \delta_a \phi^i = 0. \tag{4.58}$$

Substitute the infinitesimal transformation:

$$\sum_i \frac{\partial V}{\partial \phi^i} (T_a)^i{}_j \phi^j = 0. \tag{4.59}$$

This identity holds for every field configuration ϕ, not only at the minimum.

Now differentiate this identity with respect to ϕ^k:

$$\sum_i \frac{\partial^2 V}{\partial \phi^k \partial \phi^i} (T_a)^i{}_j \phi^j + \sum_i \frac{\partial V}{\partial \phi^i} (T_a)^i{}_k = 0. \tag{4.60}$$

Evaluate at a vacuum ϕ_0. Since ϕ_0 is a minimum,

$$\left. \frac{\partial V}{\partial \phi^i} \right|_{\phi_0} = 0. \tag{4.61}$$

The second term therefore vanishes. Define the mass matrix

$$(M^2)_{ki} = \left. \frac{\partial^2 V}{\partial \phi^k \partial \phi^i} \right|_{\phi_0}. \tag{4.62}$$

Then

$$\sum_i (M^2)_{ki} (T_a)^i{}_j \phi_0^j = 0. \tag{4.63}$$

In vector notation this is

$$M^2 \, (T_a \phi_0) = 0. \tag{4.64}$$

Now distinguish two cases.

Unbroken generator.

If T_a is unbroken, it leaves the vacuum invariant:

$$T_a \phi_0 = 0. \tag{4.65}$$

Then the equation above is trivial and gives no new massless field.

Broken generator.

If T_a is broken, then it moves the vacuum:

$$T_a \phi_0 \neq 0. \tag{4.66}$$

But the mass-matrix equation says

$$M^2 (T_a \phi_0) = 0. \tag{4.67}$$

Thus $T_a \phi_0$ is a nonzero eigenvector of the mass matrix with eigenvalue zero. A zero eigenvalue of the scalar mass matrix means a massless scalar mode. Therefore each broken generator gives a massless scalar field.

This is the mathematical form of the geometric picture. Broken generators move the vacuum along the manifold of degenerate minima. Motion along that manifold costs no potential energy, so the corresponding direction has zero curvature. Zero curvature of the potential is precisely zero mass squared.

2. The Higgs Mechanism

2.1 Difference between Higgs mechanism and Goldstone's theorem

As we have discussed in last section, Goldstone's theorem should be applied directly only to a *global* continuous symmetry. Suppose a continuous global symmetry is a true physical symmetry of the Lagrangian, but the vacuum does not respect it. Then each broken generator produces one physical massless scalar particle, called a Goldstone boson.

The Higgs mechanism uses a scalar potential with the same geometric shape, but now the symmetry is *local*. A local gauge symmetry is not an ordinary physical symmetry that relates two different states; it is a redundancy in our description of the same physical state. For this reason, Goldstone's theorem does not say that a broken gauge symmetry must produce physical massless particles.

The correct way to think about the Higgs mechanism is the following. First, temporarily ignore the gauge field. The scalar potential then has flat angular directions, and fluctuations along those directions would be Goldstone bosons in the corresponding global theory. After the symmetry is gauged, those same angular fields are no longer physical particles. They can be removed by a gauge choice and become the longitudinal polarizations of the massive gauge bosons. This is why they are called *would-be Goldstone bosons*.

Thus, when we later write a phase field such as $\theta(x)$, we are not claiming that θ is a physical massless Goldstone particle in the gauge theory. We are identifying the angular scalar mode which would have been a Goldstone boson before gauging, and then showing explicitly how the gauge field absorbs it.

In short,

$$\begin{aligned} \text{global spontaneous symmetry breaking} &\Longrightarrow \text{physical massless Goldstone bosons}, \\ \text{gauged version of the same scalar theory} &\Longrightarrow \text{would-be Goldstone modes are eaten}. \end{aligned} \tag{4.68}$$

The result is a massive gauge field. Its extra longitudinal polarization comes from the scalar mode which would have been a Goldstone boson in the ungauged global theory.

2.2 The Abelian Higgs model

We first study the simplest example: a complex scalar field coupled to a $U(1)$ gauge field. The Lagrangian is

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + (D_\mu\phi)^*(D^\mu\phi) - V(\phi), \qquad D_\mu = \partial_\mu + \mathrm{i}eA_\mu, \tag{4.69}$$

with

$$V(\phi) = -\mu^2|\phi|^2 + \frac{\lambda}{2}|\phi|^4, \qquad \mu^2 > 0, \quad \lambda > 0. \tag{4.70}$$

Equivalently, the scalar part of the Lagrangian contains

$$\mathcal{L} \supset (D_\mu\phi)^*(D^\mu\phi) + \mu^2|\phi|^2 - \frac{\lambda}{2}|\phi|^4. \tag{4.71}$$

Notice that the sign of quadratic term is positive, which means this is not the mass term yet. The Lagrangian is invariant under the local transformation

$$\phi(x) \longrightarrow e^{\mathrm{i}\alpha(x)}\phi(x), \qquad A_\mu(x) \longrightarrow A_\mu(x) - \frac{1}{e}\partial_\mu\alpha(x). \tag{4.72}$$

The reason for introducing A_μ is exactly to make derivatives compatible with a position-dependent phase. Indeed,

$$D_\mu\phi = (\partial_\mu + \mathrm{i}eA_\mu)\phi \quad \Longrightarrow \quad D_\mu\phi \longrightarrow e^{\mathrm{i}\alpha(x)}D_\mu\phi, \tag{4.73}$$

so $(D_\mu\phi)^*(D^\mu\phi)$ is gauge invariant.

The vacuum

The vacuum is found by minimizing V. Since V depends only on $|\phi|$, write $r = |\phi|$. Then

$$V(r) = -\mu^2r^2 + \frac{\lambda}{2}r^4. \tag{4.74}$$

The stationary condition is

$$\frac{\mathrm{d}V}{\mathrm{d}r} = -2\mu^2r + 2\lambda r^3 = 2r(-\mu^2 + \lambda r^2) = 0. \tag{4.75}$$

Thus either $r = 0$, or

$$r^2 = \frac{\mu^2}{\lambda}. \tag{4.76}$$

The point $r = 0$ is a local maximum, because

$$\left.\frac{\mathrm{d}^2V}{\mathrm{d}r^2}\right|_{r=0} = -2\mu^2 < 0. \tag{4.77}$$

Therefore the minima lie on the circle

$$|\phi| = v, \qquad v^2 = \frac{\mu^2}{\lambda}. \tag{4.78}$$

This circle of minima is the familiar "Mexican-hat" vacuum manifold. The radial direction costs energy, while moving around the circle costs no potential energy before the field is coupled to a gauge field.

Polar form and unitary gauge

Near the vacuum it is useful to separate radial and angular fluctuations:

$$\phi(x) = \left(v + \frac{h(x)}{\sqrt{2}}\right) e^{\mathrm{i}\theta(x)}. \tag{4.79}$$

Here $h(x)$ is the radial fluctuation, while $\theta(x)$ is the angular coordinate along the circle of classical minima. It is important not to misread this sentence: in the local $U(1)$ theory, $\theta(x)$ is not a physical Goldstone particle. Rather, it is the field that *would have been* the Goldstone boson if the same scalar potential had only a global $U(1)$ symmetry. Because the $U(1)$ symmetry is gauged here, $\theta(x)$ is a would-be Goldstone mode and can be removed by a gauge transformation.

To see this explicitly, substitute (4.79) into the covariant derivative:

$$D_\mu\phi = e^{\mathrm{i}\theta}\left[\frac{1}{\sqrt{2}}\partial_\mu h + \mathrm{i}\left(v + \frac{h}{\sqrt{2}}\right)(\partial_\mu\theta + eA_\mu)\right]. \tag{4.80}$$

The phase $e^{\mathrm{i}\theta}$ drops out of the absolute value, so

$$|D_\mu\phi|^2 = \frac{1}{2}(\partial_\mu h)(\partial^\mu h) + e^2\left(v + \frac{h}{\sqrt{2}}\right)^2 B_\mu B^\mu, \qquad B_\mu \equiv A_\mu + \frac{1}{e}\partial_\mu\theta. \tag{4.81}$$

The combination B_μ is what remains after choosing unitary gauge, where the scalar is made real:

$$\phi(x) \longrightarrow v + \frac{h(x)}{\sqrt{2}}, \qquad A_\mu(x) \longrightarrow B_\mu(x). \tag{4.82}$$

Thus the angular field θ has not disappeared mysteriously; it has become part of the gauge field.

Expanding the second term in (4.81) gives

$$e^2\left(v + \frac{h}{\sqrt{2}}\right)^2 B_\mu B^\mu = e^2v^2B_\mu B^\mu + \sqrt{2}e^2v\,hB_\mu B^\mu + \frac{e^2}{2}h^2B_\mu B^\mu. \tag{4.83}$$

The first term is a mass term for the vector field. A massive vector mass term is conventionally written as

$$\frac{1}{2}m_A^2 B_\mu B^\mu. \tag{4.84}$$

Comparing this with $e^2v^2B_\mu B^\mu$, we obtain

$$m_A^2 = 2e^2v^2. \tag{4.85}$$

The remaining terms in (4.83) are interactions between the Higgs fluctuation h and the massive vector boson.

The radial scalar mass follows from the potential. Put

$$r = v + \frac{h}{\sqrt{2}}. \tag{4.86}$$

Since $V'(v) = 0$, the leading nonconstant term is

$$V(r) = V(v) + \frac{1}{2}V''(v)\left(\frac{h}{\sqrt{2}}\right)^2 + \cdots. \tag{4.87}$$

Now

$$V''(r) = -2\mu^2 + 6\lambda r^2, \qquad V''(v) = -2\mu^2 + 6\lambda v^2 = 4\mu^2, \tag{4.88}$$

where $v^2 = \mu^2/\lambda$ has been used. Hence

$$V(r) = V(v) + \mu^2 h^2 + \cdots . \tag{4.89}$$

Because the Lagrangian contains $-V$, the quadratic scalar term is

$$-\mu^2 h^2 = -\frac{1}{2}(2\mu^2)h^2, \tag{4.90}$$

so

$$m_h^2 = 2\mu^2. \tag{4.91}$$

Counting degrees of freedom

Before the Higgs mechanism, the complex scalar has two real degrees of freedom and the massless gauge boson has two transverse polarizations:

$$2 + 2 = 4. \tag{4.92}$$

After choosing the vacuum, only the radial scalar h remains as a physical scalar. The gauge boson is now massive, so it has three polarizations:

$$1 + 3 = 4. \tag{4.93}$$

No physical degree of freedom is lost. The would-be Goldstone mode supplies the longitudinal polarization of the massive vector boson.

	scalar degrees of freedom	vector degrees of freedom	total
before	2	2	4
after	1	3	4

2.3 The same physics in Cartesian fields

The polar form makes the Higgs mechanism transparent, but it is also useful to see how the result appears if we expand the complex scalar into two real fields:

$$\phi(x) = v + \frac{1}{\sqrt{2}}\big(\phi_1(x) + \mathrm{i}\phi_2(x)\big), \qquad \phi_1, \phi_2 \in \mathbb{R}. \tag{4.94}$$

Here ϕ_1 is the radial fluctuation and ϕ_2 is the angular fluctuation to leading order.

To derive the quadratic part of the potential, first compute

$$|\phi|^2 = \left(v + \frac{\phi_1}{\sqrt{2}}\right)^2 + \left(\frac{\phi_2}{\sqrt{2}}\right)^2 = v^2 + \sqrt{2}v\phi_1 + \frac{1}{2}(\phi_1^2 + \phi_2^2). \tag{4.95}$$

It is useful to write this as

$$|\phi|^2 = v^2 + \Delta, \qquad \Delta = \sqrt{2}v\phi_1 + \frac{1}{2}(\phi_1^2 + \phi_2^2). \tag{4.96}$$

Substituting this into $V = -\mu^2|\phi|^2 + \frac{\lambda}{2}|\phi|^4$ gives

$$\begin{aligned} V &= -\mu^2(v^2+\Delta) + \frac{\lambda}{2}(v^2+\Delta)^2 \\ &= V(v) + (-\mu^2 + \lambda v^2)\Delta + \frac{\lambda}{2}\Delta^2. \end{aligned} \tag{4.97}$$

The coefficient of Δ vanishes because the vacuum satisfies $v^2 = \mu^2/\lambda$. Therefore there is no linear term, and the quadratic part comes only from Δ^2. To second order in the fluctuating fields,

$$\Delta^2 = \left(\sqrt{2}v\phi_1\right)^2 + \text{terms of cubic or higher order} = 2v^2\phi_1^2 + \cdots . \tag{4.98}$$

Using $v^2 = \mu^2/\lambda$, the potential therefore expands as

$$V(\phi) = V(v) + \mu^2\phi_1^2 + \text{interaction terms.} \tag{4.99}$$

Therefore

$$-V(\phi) = -V(v) - \mu^2\phi_1^2 + \text{interaction terms.} \tag{4.100}$$

This gives

$$m_{\phi_1}^2 = 2\mu^2, \qquad m_{\phi_2}^2 = 0 \tag{4.101}$$

before gauge fixing. The absence of a ϕ_2^2 term is the usual Goldstone-boson result for the scalar potential. In a gauge theory this field will mix with the gauge field and is not a physical massless particle.

Now expand the covariant derivative:

$$\begin{aligned} D_\mu\phi &= \partial_\mu\phi + \mathrm{i}eA_\mu\phi \\ &= \frac{1}{\sqrt{2}}\left(\partial_\mu\phi_1 + \mathrm{i}\partial_\mu\phi_2\right) + \mathrm{i}evA_\mu + \frac{\mathrm{i}e}{\sqrt{2}}A_\mu(\phi_1 + \mathrm{i}\phi_2). \end{aligned} \tag{4.102}$$

Keeping only terms quadratic in the fields gives

$$|D_\mu\phi|^2 = \frac{1}{2}(\partial_\mu\phi_1)^2 + \frac{1}{2}(\partial_\mu\phi_2)^2 + \sqrt{2}ev\, A^\mu\partial_\mu\phi_2 + e^2v^2A_\mu A^\mu + \cdots . \tag{4.103}$$

The last term is the same gauge-boson mass term as before:

$$e^2v^2A_\mu A^\mu = \frac{1}{2}m_A^2A_\mu A^\mu, \qquad m_A^2 = 2e^2v^2. \tag{4.104}$$

The mixed term

$$\sqrt{2}ev\, A^\mu\partial_\mu\phi_2 \tag{4.105}$$

is the Cartesian-field signal that ϕ_2 is not an independent physical particle. In unitary gauge it is removed. In a covariant R_ξ gauge it is cancelled by a gauge-fixing term, leaving a gauge-dependent unphysical would-be Goldstone field whose effects cancel from physical amplitudes.

2.4 What the propagator is telling us

Now we shall top for a moment and discuss how “the gauge field eats the would-be Goldstone mode” . And what really shows up in the propagator of the gauge field.

Recall the degree-of-freedom count:

$$\text{massless vector: 2 polarizations,} \qquad \text{massive vector: 3 polarizations.} \tag{4.106}$$

The new third polarization is the longitudinal polarization. The question is: where does it come from? The answer is that it comes from the angular scalar mode, namely the would-be Goldstone field.

In unitary gauge the would-be Goldstone field has been removed from the scalar field, so it is hidden inside the gauge field. The massive-vector propagator then takes the form

$$D^{\text{unitary}}_{\mu\nu}(k) = \frac{-\mathrm{i}}{k^2 - m_A^2 + \mathrm{i}\epsilon}\left(g_{\mu\nu} - \frac{k_\mu k_\nu}{m_A^2}\right). \tag{4.107}$$

The term $k_\mu k_\nu / m_A^2$ is the part associated with the longitudinal polarization. Therefore this propagator is the unitary-gauge way of saying: the vector field now contains a longitudinal mode.

The same physics can be seen before choosing unitary gauge. In the Cartesian expansion, the quadratic Lagrangian contains the mixing term

$$\sqrt{2}ev\, A^\mu \partial_\mu \phi_2. \tag{4.108}$$

Using $m_A = \sqrt{2}ev$, this is

$$m_A A^\mu \partial_\mu \phi_2. \tag{4.109}$$

The derivative gives a momentum factor k_μ, so the mixing between A_μ and the would-be Goldstone field carries a factor schematically like $m_A k_\mu$. This is why a would-be Goldstone exchange naturally produces terms proportional to $k_\mu k_\nu$, exactly the tensor structure needed for the longitudinal part of a vector propagator.

Schematically, the gauge-boson mass insertion gives a contribution

$$\mathrm{i}m_A^2 g_{\mu\nu}. \tag{4.110}$$

The exchange of the would-be Goldstone mode gives a longitudinal contribution of the form

$$m_A k_\mu \frac{\mathrm{i}}{k^2}(-m_A k_\nu) = -\mathrm{i}m_A^2 \frac{k_\mu k_\nu}{k^2}. \tag{4.111}$$

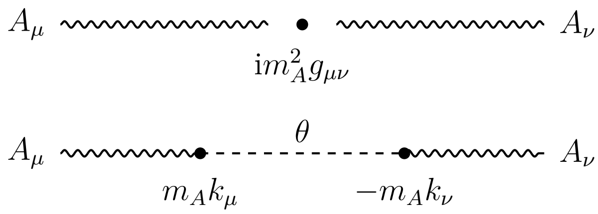

Figure 4.3: The mass insertion and the would-be Goldstone exchange combine to produce the longitudinal structure of the massive gauge boson. This is a schematic diagram; the precise signs depend on the gauge-fixing convention.

Together they form the transverse structure

$$\mathrm{i}m_A^2\left(g_{\mu\nu} - \frac{k_\mu k_\nu}{k^2}\right). \tag{4.112}$$

This equation should be read only as a schematic explanation, not as a complete propagator formula. The lesson is simple: the same scalar mode that disappears in unitary gauge is precisely the mode that supplies the longitudinal polarization of the massive gauge boson. Thus the phrase "the gauge boson eats the Goldstone boson" means:

$$A_\mu \text{ with 2 polarizations} + 1 \text{ would-be Goldstone mode} \implies \text{massive } A_\mu \text{ with 3 polarizations}. \tag{4.113}$$

2.5 $SU(2)$ gauge symmetry breaking

The non-Abelian version uses a complex scalar doublet

$$\Phi = \begin{pmatrix} \Phi_1 \\ \Phi_2 \end{pmatrix}, \qquad \Phi_1, \Phi_2 \in \mathbb{C}, \tag{4.114}$$

coupled to an $SU(2)$ gauge field. The Lagrangian is

$$\mathcal{L} = -\frac{1}{4} G^a_{\mu\nu} G^{a\mu\nu} + (D_\mu \Phi)^\dagger (D^\mu \Phi) - V(\Phi), \tag{4.115}$$

where

$$V(\Phi) = -\mu^2 \Phi^\dagger \Phi + \lambda (\Phi^\dagger \Phi)^2, \qquad \mu^2 > 0, \quad \lambda > 0, \tag{4.116}$$

and

$$D_\mu = \partial_\mu - \mathrm{i} g \frac{\sigma^a}{2} W^a_\mu. \tag{4.117}$$

The σ^a are the Pauli matrices, and repeated adjoint indices $a = 1, 2, 3$ are summed.

The potential depends only on $\Phi^\dagger \Phi$. Let

$$\rho^2 = \Phi^\dagger \Phi. \tag{4.118}$$

Then

$$V(\rho) = -\mu^2 \rho^2 + \lambda \rho^4. \tag{4.119}$$

The minimum satisfies

$$\frac{\mathrm{d}V}{\mathrm{d}\rho} = -2\mu^2 \rho + 4\lambda \rho^3 = 2\rho(-\mu^2 + 2\lambda \rho^2) = 0. \tag{4.120}$$

Thus the nonzero vacuum satisfies

$$\Phi^\dagger \Phi = \rho^2 = \frac{\mu^2}{2\lambda}. \tag{4.121}$$

It is conventional to write

$$\langle \Phi \rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ v \end{pmatrix}, \qquad v^2 = \frac{\mu^2}{\lambda}. \tag{4.122}$$

Indeed,

$$\langle \Phi \rangle^\dagger \langle \Phi \rangle = \frac{v^2}{2} = \frac{\mu^2}{2\lambda}. \tag{4.123}$$

Notice the dimensions: $\Phi^\dagger \Phi$ has dimension two, so it is proportional to v^2, not to v.

Which generators are broken?

For this pure $SU(2)$ theory, all three generators are broken by the vacuum. Acting with the Pauli matrices gives

$$\sigma^1 \begin{pmatrix} 0 \\ v \end{pmatrix} = \begin{pmatrix} v \\ 0 \end{pmatrix}, \qquad \sigma^2 \begin{pmatrix} 0 \\ v \end{pmatrix} = \begin{pmatrix} -\mathrm{i}v \\ 0 \end{pmatrix}, \qquad \sigma^3 \begin{pmatrix} 0 \\ v \end{pmatrix} = \begin{pmatrix} 0 \\ -v \end{pmatrix}. \tag{4.124}$$

None of these is zero, so the vacuum is moved by every generator. Hence the three would-be Goldstone fields are absorbed by the three $SU(2)$ gauge bosons.

This statement is specific to a pure $SU(2)$ gauge theory with one scalar doublet. In the Standard Model the gauge group is $SU(2)_L \times U(1)_Y$, and one linear combination remains unbroken; that unbroken combination is electromagnetism.

Unitary gauge and the gauge-boson masses

The most general fluctuation around the chosen vacuum may be written as

$$\Phi(x) = \exp\left[\frac{\mathrm{i}}{2}\sigma^a\theta^a(x)\right]\frac{1}{\sqrt{2}}\begin{pmatrix}0\\ v+h(x)\end{pmatrix}. \tag{4.125}$$

The three fields $\theta^a(x)$ are the would-be Goldstone fields. An $SU(2)$ gauge transformation can remove them, leaving unitary gauge:

$$\Phi(x) = \frac{1}{\sqrt{2}}\begin{pmatrix}0\\ v+h(x)\end{pmatrix}. \tag{4.126}$$

In matrix notation,

$$W_\mu = W_\mu^a\frac{\sigma^a}{2}. \tag{4.127}$$

Under a local $SU(2)$ transformation $U(x)$, the gauge field transforms as

$$W_\mu \longrightarrow UW_\mu U^\dagger - \frac{\mathrm{i}}{g}U\partial_\mu U^\dagger, \tag{4.128}$$

with the sign matching the convention

$$D_\mu = \partial_\mu - \mathrm{i}gW_\mu. \tag{4.129}$$

Now compute the covariant derivative in unitary gauge. Since

$$\sigma^1\begin{pmatrix}0\\ v+h\end{pmatrix} = \begin{pmatrix}v+h\\ 0\end{pmatrix}, \quad \sigma^2\begin{pmatrix}0\\ v+h\end{pmatrix} = \begin{pmatrix}-\mathrm{i}(v+h)\\ 0\end{pmatrix}, \quad \sigma^3\begin{pmatrix}0\\ v+h\end{pmatrix} = \begin{pmatrix}0\\ -(v+h)\end{pmatrix}, \tag{4.130}$$

we find

$$D_\mu\Phi = \frac{1}{\sqrt{2}}\begin{pmatrix}-\frac{\mathrm{i}g}{2}(v+h)(W_\mu^1 - \mathrm{i}W_\mu^2)\\ \partial_\mu h + \frac{\mathrm{i}g}{2}(v+h)W_\mu^3\end{pmatrix}. \tag{4.131}$$

Taking the Hermitian norm gives

$$(D_\mu\Phi)^\dagger(D^\mu\Phi) = \frac{1}{2}(\partial_\mu h)(\partial^\mu h) + \frac{g^2}{8}(v+h)^2W_\mu^aW^{a\mu}. \tag{4.132}$$

The gauge-boson mass term is therefore

$$\frac{g^2v^2}{8}W_\mu^aW^{a\mu}. \tag{4.133}$$

Since the canonical mass term for each real vector boson is

$$\frac{1}{2}m_W^2W_\mu^aW^{a\mu}, \tag{4.134}$$

we obtain

$$m_W^2 = \frac{g^2v^2}{4}. \tag{4.135}$$

All three gauge bosons have the same mass because the original theory has an $SU(2)$ symmetry and the scalar representation treats the three generators symmetrically after all of them are broken.

Degree-of-freedom count for the $SU(2)$ model

The complex doublet contains four real scalar degrees of freedom. Before the Higgs mechanism, the three massless gauge bosons each have two transverse polarizations:

$$4 + 3 \times 2 = 10. \tag{4.136}$$

After the Higgs mechanism, one scalar h remains, and the three gauge bosons are massive, each with three polarizations:

$$1 + 3 \times 3 = 10. \tag{4.137}$$

Again the number of physical degrees of freedom is unchanged. The three would-be Goldstone modes have become the longitudinal polarizations of the three massive gauge bosons.

	scalar degrees of freedom	vector degrees of freedom	total
before	4	$3 \times 2 = 6$	10
after	1	$3 \times 3 = 9$	10